

Streaming Success: A Data-Driven Analysis of Netflix Content Distribution and Evolution

George Mason University
AIT-580-009 | Prof.Dr.Harry Foxwell, Prof.Raymond.P.Smith

Snehita Vattamwar
Department of Applied Information Technology
George Mason University
Fairfax, Virginia
svattamw@gmu.edu

ABSTRACT

Imagine the time where your favorite shows and movies might come from any location on the planet—this is the world Netflix is creating. In our study we examine how Netflix has evolved into a global entertainment destination, giving a diverse choice of episodes and movies to consumers all around the world. We sifted through a big amount of Netflix data, using unique algorithms to determine whether they are offering more TV series currently than they were previously, as well as how their selection vary by country.

Our investigation showed that Netflix isn't just about Hollywood anymore; it's become a mosaic of global stories, with more TV series popping up on the menu. The reason? Viewers everywhere love to binge-watch series and enjoy tales from different cultures. This research could be a treasure map for filmmakers and producers, pointing them to what viewers around the world might want to see next on their screens.

Keywords—Netflix, TV Series, Movies, Distribution, Trends, Countries, Genres.

I. INTRODUCTION

Since the 1980s, when scientists began to sound the alarm, streaming platforms have reshaped the entertainment sector, fundamentally changing our interaction with media and culture. At the vanguard of this revolution, Netflix has evolved from a simple DVD rental business into a major player in the era of digital streaming. This study aims to explore how Netflix has steered and shaped the transformation of media consumption and its wider effects on global cultural exchanges and production methodologies. The United Nations has warned that similar to the global need for managing our environmental impact to combat climate change, the media industry must also adjust to the rapidly changing digital landscape to maintain cultural and economic sustainability [1].

Netflix's rise has mirrored the growth of the internet and the broader digital revolution. The company's transition from distributing DVDs to streaming content online in 2007 was a critical juncture, leveraging technological progress to revolutionize how entertainment is accessed and experienced [2]. This shift is akin to global initiatives aimed at tackling climate issues—both demand innovation and the adoption of new sustainable models [3].

To address the research question regarding the evolution of Netflix's content distribution strategies, it's crucial to assess the company's impact on consumer habits and its role in mirroring and molding cultural trends. Netflix's vast array of programming and original content has established a new standard in media production and consumption, promoting a globalized and interconnected cultural realm. The influence of Netflix on global entertainment and culture mirrors the comprehensive, integrated efforts needed to tackle global warming, with the company's strategies serving as a model for adaptation and transformation [4].

This study delves into three key aspects of Netflix's content strategy and its development within the streaming industry:

a. Evolution of content distribution across countries over time:

Netflix's approach to distributing content across various nations has dynamically evolved, shaped by globalization and the broader access to internet worldwide. This research examines Netflix's expansion in offering diverse content tailored to different cultural preferences and consumer demands across regions. The objective is to outline how content availability trends over time and across geographies, assessing

if there has been a strategic pivot in distribution towards emerging markets and regions with growing content production capabilities.

b. Patterns of similarity across movies and TV series based on text attributes:

This paper investigates the similarities across Netflix's extensive content library by analyzing metadata like names of actors, directors, and genres. Employing text analytics and natural language processing, the study seeks to identify thematic groupings and specific content niches that Netflix targets. This analysis aims to provide insights into Netflix's content curation strategy, reflecting possibly on wider industry tendencies concerning viewer preferences and overall content strategies.

c. Trends in the emphasis on developing TV series versus movies:

There has been a noticeable shift in consumer preferences from traditional movies to more immersive TV series in recent years. This section explores whether Netflix's content creation strategies have adapted to this change. By reviewing the release frequencies and comparing the volume of TV series to movies produced over time, the study will explore strategic shifts and priorities within Netflix's content development efforts. This analysis will aid in understanding how Netflix positions itself within a competitive streaming landscape where serialized content is crucial for maintaining viewer engagement and sustaining subscriber loyalty.

Overall, this paper aims to provide a detailed examination of how strategic choices made by a prominent streaming service like Netflix influence and adapt to global media consumption patterns. The findings are intended to enhance academic perspectives on digital media dynamics and provide actionable insights for industry stakeholders in navigating the digital content environment.

II. RELATED WORK

The rapid expansion of over-the-top (OTT) video platforms such as Netflix has caught the interest of academics seeking to understand user behavior, platform design, and content distribution tactics. Several studies have investigated various elements of Netflix and comparable platforms, yielding significant information on user preferences, platform problems, and content recommendation algorithms.

One notable research effort by Adhikari et al. [5] delves into the architecture and performance of Netflix and Hulu, two leading OTT content providers in the US and Canada. Through

extensive measurement studies, the authors uncover the system architectures, service strategies, and content distribution network (CDN) utilization of Netflix and Hulu. They highlight similarities and differences between the two platforms, shedding light on CDN selection strategies and their impact on user experience.

In a related study, Kamarudin et al. [6] focus on understanding the viewing interests of Asian audiences on Netflix. Analyzing data from five Southeast Asian countries, the authors employ decision support system methodologies to predict and analyze the most-watched content in each country. Their research underscores the importance of localized content preferences and the potential for adaptive content recommendation strategies based on regional viewing patterns.

Another research direction explores the analysis and visualization of Netflix content to uncover trends and patterns in viewer preferences. Devashree et al. [7] conduct data analysis and visualization on Netflix shows and movies, emphasizing genre popularity and viewer demographics. By analyzing release years, genres, and ratings from databases like tmdb and imdb, the authors provide insights into content popularity and audience preferences, aiding content recommendation strategies.

Overall, these studies contribute to a comprehensive understanding of user behavior, platform dynamics, and content distribution strategies in the OTT landscape. By leveraging data analysis, visualization techniques, and predictive modeling, researchers aim to optimize content delivery, enhance user engagement, and drive business growth for OTT platforms like Netflix.

III. DATASET

The dataset used in this study was obtained from Data.world [8] and includes an extensive array of movies and television series from 1980 to 2020. It contains 7788 entries, demonstrating its significant volume. The characteristics of the dataset are detailed in Table 1, which lists the attributes of each film or series along with their data types. This dataset is carefully organized to include all measurement levels: nominal, ordinal, interval, and ratio, providing a strong foundation for analysis.

This dataset provides a comprehensive overview of the entertainment industry over forty years, reflecting changes in genre, audience preferences, and filmmaking techniques. Its variety enables researchers to investigate numerous topics, such as the development of cinematography and the effects of cultural transformations on film and television. With a range of data types, from categorical labels to numerical values, the

dataset supports diverse analyses, from qualitative thematic studies to quantitative statistical evaluations. This resource not only serves as a reflection of historical trends in media but also as a crucial tool for forecasting future trends in media consumption and production.

Table 1:

Dataset	
Attribute	Data Type
Show_id	Nominal
Type	Nominal
Title	Nominal
Director	Nominal
Cast	Nominal
Country	Nominal
Date_added	Ordinal
Release_year	Interval
Rating	Ordinal
Duration	Ratio
Genres	Nominal
Description	Nominal

IV. METHODOLOGY AND TECHNOLOGICAL FRAMEWORK

This study on Netflix content leverages a multifaceted approach that incorporates data manipulation, statistical analysis, machine learning, and cloud-based technologies to thoroughly explore the posed research queries. The methodology and tools are detailed as follows:

A. Data Preparation and Cleaning:

1. Data Loading and Cleaning with Python:

- *Python Environment*: Initial data importation will be handled through Python scripts, employing libraries such as **'pandas'** for dataset manipulation. The **'read_csv()'** function will be primarily used to load data from CSV files.

- *Data Cleaning*: For cleaning operations, libraries like **'pandas'** and **'numpy'** will be utilized to manage issues such as missing values and duplicates, and to perform data normalization and categorical variable encoding as needed.

2. AWS for Data Management and Storage:

- *AWS S3*: Once cleaned, data will be stored in AWS S3 buckets to facilitate durability and accessibility throughout different phases of the project.

- *AWS Glue DataBrew*: This tool will further assist in data cleaning and preparation via its user-friendly visual interface, which allows manipulation of data directly within S3 without requiring additional coding.

B. Text Analysis and Similarity Measurement:

1. Text Processing Using Python:

- *Employing 'tm' package*: The **'tm'** package in R will be used for constructing a document-term matrix, essential for subsequent text analysis tasks.

- *Cosine Similarity*: The **'proxy'** package will be employed to compute cosine similarity among textual elements to identify relationships among various Netflix content.

2. Advanced Text Analysis with Python:

- *Natural Language Processing with 'spaCy'*: **'spaCy'** will be utilized for advanced text processing capabilities such as tokenization, part-of-speech tagging, and named entity recognition.

- *Vectorization with 'CountVectorizer'*: This function from the **'sklearn'** library will help in converting text data into a matrix of token counts, aiding in the machine learning process.

- *Sentiment Analysis*: Sentiment analysis will be performed using Python libraries like **'nltk'** or **'textblob'** to assess the emotional undertones of content descriptions.

C. Exploratory Data Analysis (EDA) and Visualization:

1. Statistical Analysis and Visualization using R and Python:

- *RStudio*: R will be used for deriving summary statistics and preliminary visualizations like histograms and box plots via the **'ggplot2'** library.

- *Visualization with Python*: Tools such as **'matplotlib'** and **'seaborn'** will be utilized for more

comprehensive data visualization tasks to examine data trends and correlations.

D. SQL and AWS for Data Queries and Management:

1. SQL Operations using AWS RDS:

- Complex SQL queries will be executed to analyze structured data, with AWS RDS supporting scalable and efficient query handling.

2. Data Visualization and Reporting:

- *Enhanced Visualization with Python:* Python's sophisticated visualization libraries will be used to produce interactive graphics and charts for presenting findings.
 - *AWS QuickSight:* This tool will be employed for dashboard creation and insight dissemination among stakeholders, integrating data from AWS S3 and RDS.

Tools and Software :

1. *Python and RStudio:* Main environments for data handling, analysis, and visualization.
2. *AWS S3 and AWS Glue DataBrew:* Employed for data storage and preparation.
3. *AWS RDS:* Facilitates the execution of SQL queries for structured data analysis.
4. *Text Analysis Tools:* spaCy, CountVectorizer, and various sentiment analysis tools in Python to enhance text analysis capabilities.

By utilizing these tools and technologies, the study aims to employ sophisticated data analysis methods and a robust cloud infrastructure to deliver a comprehensive analysis of Netflix's content strategies and audience interactions. This approach ensures scalability for handling extensive datasets and the advanced analytical capabilities necessary for deep insights.

V. RESULTS AND ANALYSIS

After the first step of the proposed approach, Figures 1a and 1b display the first five rows of the dataset, before and after data cleaning and feature extraction. In data cleaning, we replaced the attribute 'director' with null values with the name 'Unknown' and dropped rows with missing values in other columns. The columns 'date_added' and 'release_year' are formatted to pd.datetime, and in this process, we updated the dataset with two additional columns, namely, 'month_added' and 'day_added'.

1A: Data loading and cleaning with python

Figure 1a:

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, film...
1	s2	TV Show	Blood & Water	NaN	Amia Qamata, Khosi Ngema, Gail Mabasa, Thabani...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town L...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotsis, Samuel Jouy, Nabil...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alan K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train L...

Figure 1b:

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	month_added	day	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, film...	September	25
1	s2	TV Show	Blood & Water	Unknown	Amia Qamata, Khosi Ngema, Gail Mabasa, Thabani...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town L...	September	24
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotsis, Samuel Jouy, Nabil...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	September	24
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go	September	24

1B: Connecting to AWS S3 and AWS Glue DataBrew

In the second step, we created an AWS S3 bucket as shown in Figure 2a and named the S3 bucket '580ait'. We then imported the dataset 'UpdatedNetflix.csv' into the S3 bucket, as shown in Figure 2b. To view the summary statistics and basic visualizations of the dataset present in the S3 bucket, we created a project using AWS Glue DataBrew, as shown in Figure 2, and depicted the statistics of the dataset.

Figure 2a:

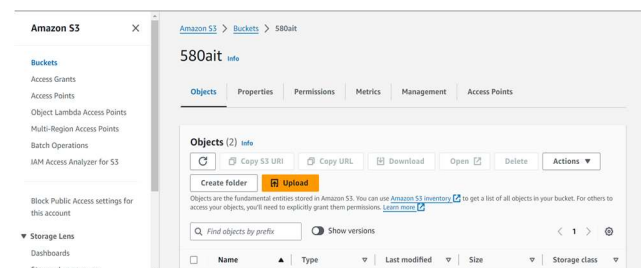


Figure 2b:

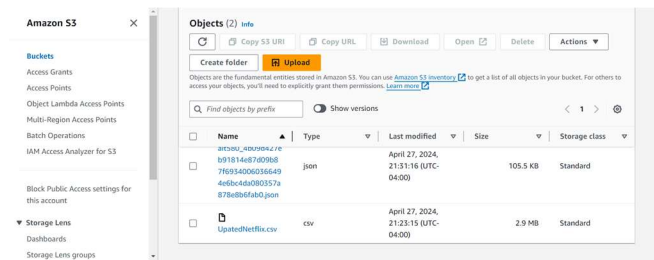


Figure 4b:

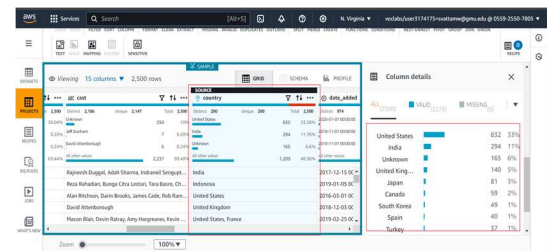
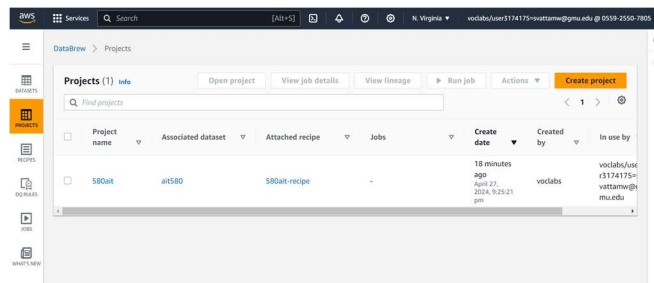


Figure 3:



Using AWS S3 and AWS Glue DataBrew, an overview of the country distribution by frequency is shown in Figure 4a. 35% of shows and movies originate from the United States, as shown with 852 records in Figure 4b. The next most common country of origin is India, with 11% and 294 records.

Figure 5 presents a statistical analysis of a dataset, showing that it has high cardinality with each row being unique, indicating a dataset that is varied and free from duplicates. The presence of outliers is noted, with 1% of the data significantly deviating from the average, as these values have a z-score exceeding the usual limit of ± 3 , totaling 98 unique outlier values. Furthermore, the dataset is noted for its completeness, with no missing values, implying it is well-organized and primed for analysis without the requirement to fill in any missing information.

Figure 4a:

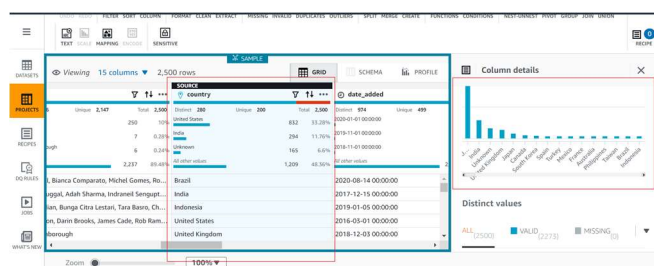
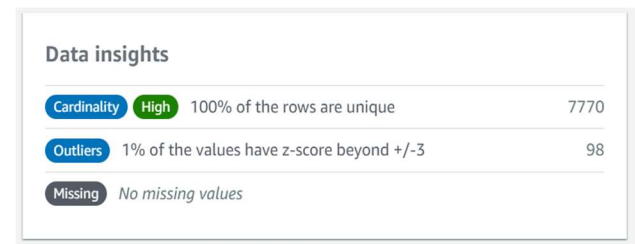


Figure 5:



Using Glue DataBrew, providing summary statistics of minimum, first quartile, median, mean, third quartile, maximum for attributes like duration, month_added, day. Figure 6a, displays a correlation matrix heatmap, which reveals the correlation coefficients among various variables in a dataset. The intensity of the colors and the numerical values in the boxes reflect the strength and direction of each relationship, with 1 indicating a perfect positive correlation and -1 a perfect negative correlation. There is a slight negative correlation between the variables 'duration' and 'month_added,' indicating that an increase in one is associated with a minor decrease in the other. The 'day' variable shows negligible linear correlation with 'duration' and a small positive correlation with 'month_added.'

Figure 6b, displays boxplot diagram analyzing the distribution of values for the variables 'duration,' 'month_added,' and 'day.' Boxplots illustrate the median, quartiles, and potential outliers within data distributions. For 'duration,' the median is positioned near the lower quartile, indicating a distribution that is skewed with some higher values. The distributions of 'month_added' and 'day' appear more uniform, though without specific scale details, assessing their precise characteristics is challenging. Together, these images offer a comprehensive visual overview of the statistical relationships and value distributions within the dataset.

Figure 6a:

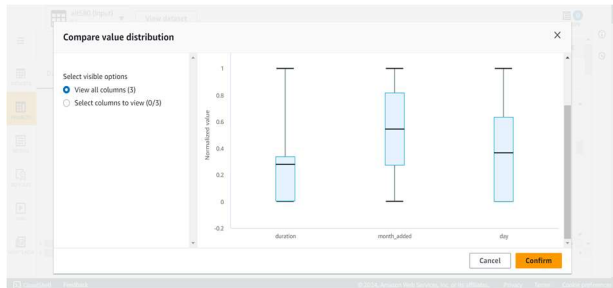


Figure 6b:



B: Text Analysis and Similarity Measurement using Python (Jupyter IDE):

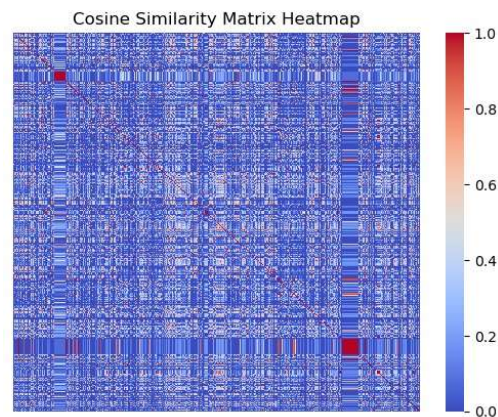
In Step B of proposed approach, providing text analysis within the 'listed_in' column of a 'netflix' dataset, focusing on the initial 500 records. Using 'TfidfVectorizer' from scikit-learn to convert the text into a TF-IDF matrix, which quantifies the relevance of words within the text relative to the dataset as a whole. After transforming the text, the code employs the 'cosine_similarity' function to create a 500x500 matrix representing the cosine similarities between pairs of TF-IDF vectors. Figure 7 is the similarity matrix facilitates the discovery of patterns and relationships among the entries, which is advantageous for content-based filtering, making recommendations, or clustering similar genres/categories within the dataset. Then displaying a cosine similarity matrix heatmap as shown in figure 8, illustrating the pairwise similarity among items, likely resulting from a text analysis using a TF-IDF framework. The diagonal, prominently colored in red, indicates a perfect similarity score of 1 for each item with itself, a typical characteristic of such matrices. Around this diagonal, the transition in colors from blue to red represents the spectrum of similarity scores, where blue denotes low similarity (scores near 0) and red signifies high

similarity (scores close to 1). The appearance of red squares and clusters apart from the diagonal implies the existence of subsets of items that are highly similar to each other, which might suggest groupings based on common content or themes. Predominantly, the matrix is covered in blue, showing that the majority of items are distinctly different from each other. Due to the matrix's dense visual presentation, it is difficult to identify specific entries, highlighting the potential need for a more interactive or refined analytical method to derive practical insights.

Figure 7:

```
similarity_matrix
Out[65]: array([[1., 0., 0., ..., 0., 0.,
0., 1., 0.63722856, ..., 0., 0.27279337,
0., 1.,
0., 0.63722856, 1., ..., 0., 0.10745652,
0., 1.,
...,
0., 0., 0., ..., 1., 0.,
1., 1.,
0., 0.27279337, 0.10745652, ..., 0., 1.,
0., 1.,
0., 0., 0., ..., 1., 0.,
1.]])
```

Figure 8:



Using spacy helps clarifying the types of entities present within a given text corpus by entity extraction, Figure 9 displaying a list of named entities extracted from a data, each tagged with their corresponding entity types. For instance, names such as "Kirsten Johnson" and "Mehdi" are identified as "PERSON," suggesting they represent individuals. Places like "New Orleans" and "Earth" are categorized under "GPE" (geo-political entity) and "LOC" (location) respectively. Similarly, "Ghana" and "India" are accurately labeled as "GPE," while "American" is categorized as "NORP," indicating a national, religious, or political group. Cities like

"Bangkok" and "Bengaluru" are also rightly identified as "GPE." Additionally, "Tet  " is tagged as a "PERSON."

Figure 9:

```
Kirsten Johnson PERSON
Mehdi ORG
the Orleans Justice Center ORG
New Orleans GPE
India GPE
Equestria PERSON
Earth LOC
Pegasi GPE
Unicorns ORG
Ghana GPE
American NORP
10-week DATE
U.K. GPE
Anti-Mafia WORK_OF_ART
Bangkok GPE
Tet   PERSON
Bengaluru GPE
four CARDINAL
```

Using one more method, **CountVectorizer** from scikit-learn to analyze a dataset containing descriptions, extracting words while omitting common English stop words. It then transforms these descriptions into a numerical array where the occurrences of each word across all descriptions are counted. The output from figure 10 reveals that the most frequently occurring words are terms such as 'life', 'young', 'new', 'family', and 'world', suggesting themes centered around personal and social dynamics and utilizing the **SentimentIntensityAnalyzer** from nltk's vader module to assess the sentiment of each description in a dataset. By downloading the necessary vader_lexicon, the analyzer computes sentiment scores for each text, identifying the percentages of negative, neutral, and positive sentiments, along with a compound score that summarizes the overall sentiment. For instance, the first description in the dataset is analyzed to have more neutral content with a slight negative bias and minimal positive elements, indicated by a compound score of -0.296 as figure 11.

Figure 10:

```
[('life', 845), ('young', 729), ('new', 709), ('family', 635), ('world', 570), ('man', 538), ('love', 504), ('woman', 486), ('friends', 445), ('series', 396)]
```

Figure 11:

```
{'neg': 0.123, 'neu': 0.791, 'pos': 0.085, 'compound': -0.296}
```

Step C: Exploratory Data Analysis (EDA) and Visualization using R Language

In this step using R language, Figure12 shows a structured summary of a dataset using R programming, detailing the properties of various columns. Each block represents a column in the dataset, which includes 'show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', 'description', 'month_added', and 'day'.

Figure 12:

```
show_id      type      title
Length:8790  Length:8790 Length:8790
Class :character  Class :character  Class :character
Mode :character  Mode :character  Mode :character

director     cast      country
Length:8790  Length:8790 Length:8790
Class :character  Class :character  Class :character
Mode :character  Mode :character  Mode :character

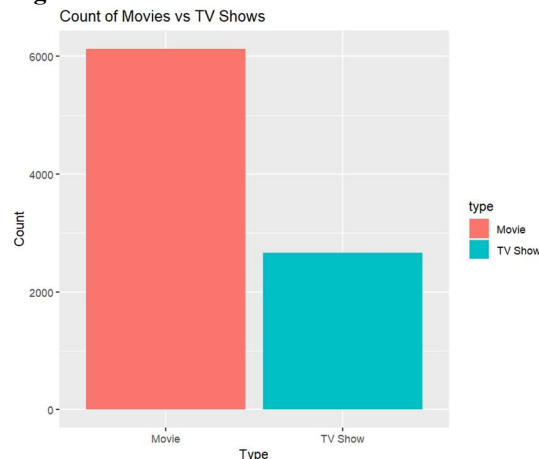
date_added   release_year  rating  duration
Length:8790  Min. :1925    Length:8790 Length:8790
Class :character  1st Qu.:2013  Class :character  Class :character
Mode :character  Median :2017  Mode :character  Mode :character
                  Mean :2014
                  3rd Qu.:2019
                  Max. :2021

listed_in    description  month_added  day
Length:8790  Length:8790  Length:8790  Min. : 1.0
Class :character  Class :character  Class :character  1st Qu.: 1.0
Mode :character  Mode :character  Mode :character  Median :13.0
                  Mean :12.5
                  3rd Qu.:20.0
                  Max. :31.0
```

> |

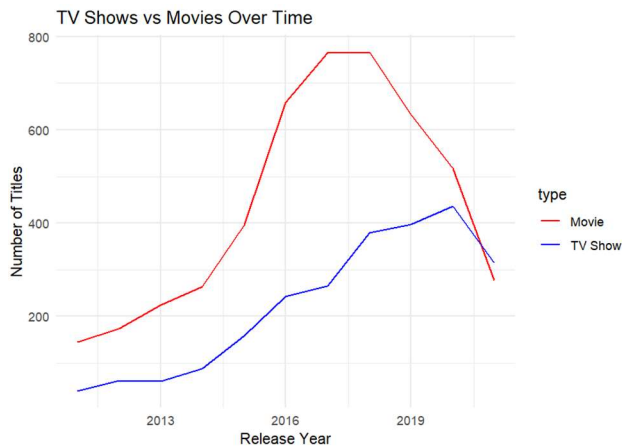
The figure 13 shows a bar chart titled "Count of Movies vs TV Shows." It compares the number of movies to the number of TV shows in a dataset. The each bar represents movies or TV shows, the one colored in red movies, is significantly more than the bar for TV shows, which is colored in teal. The graph suggests that there are more movies than TV shows in the dataset, with movies numbering more than 6000 and TV shows numbering around 2400. The y-axis represents the count of each type, while the x-axis is represents 'Type' with two categories: 'Movie' and 'TV Show'.

Figure 13:



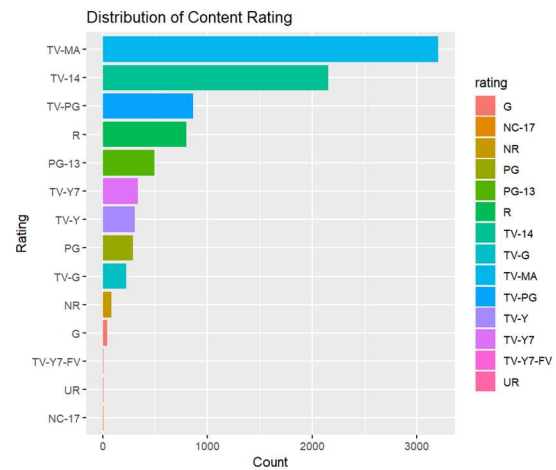
The figure 14 shows a line graph titled "TV Shows vs Movies over Time," which compares the number of TV show and movie titles released over a range of years. The x-axis and y-axis indicates the release year and number of titles. The red line represents movies, while the blue line represents TV shows. From the graph, it is clear that the number of movies saw a significant increase, peaking at around the year 2018 before experiencing a sharp decline. In the opposite, the number of TV shows has been steadily increasing over the years. The trend for TV shows does not show a sharp peak as seen with movies, rather it exhibits a more gradual rise. The last visible data point suggests the graph may be cut off, so recent trends for TV shows cannot be determined conclusively. Overall, the graph illustrates the changing dynamics between movies and TV shows in terms of release volumes over the years covered by the dataset.

Figure 14:



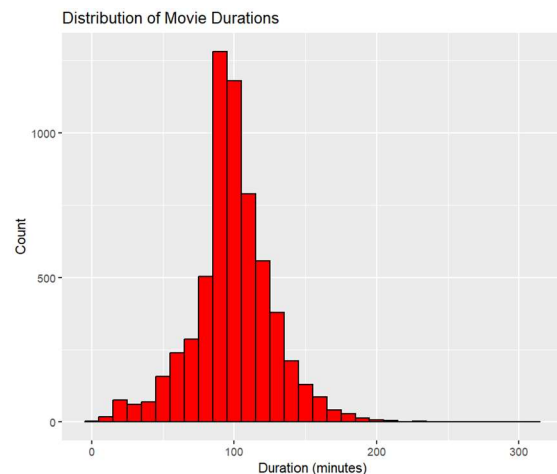
The figure 15 depicts a horizontal bar chart titled "Distribution of Content Rating," which displays the frequency of various content ratings in dataset. Each bar represents a different content rating category. The dataset includes ratings such as G, NC-17, NR (Not Rated), PG, PG-13, R, TV-14, TV-G (General Audience), TV-MA (Mature Audience), TV-PG (Parental Guidance), TV-Y (All Children), TV-Y7 (Older Children), TV-Y7-FV (Fantasy Violence), and UR (Unrated). The most of the content represents the 'TV-MA' rating. Other most represented categories are 'TV-14' and 'TV-PG'. Ratings such as 'G' and 'NC-17' has lowest content ratings.

Figure 15:



The figure 16 depicts a histogram titled "Distribution of Movie Durations," displaying the frequency of movie durations in minutes. The x-axis and y-axis represents the duration of movies in minutes and count. The graph distribution appears to be centered around a duration which could be roughly 100 minutes. The shape of the distribution is fairly symmetrical and bell-shaped, indicating that most of the movies in the dataset have durations close to this central value, with fewer movies having much shorter or much longer durations. The decline in frequency as durations get longer suggests that very long movies are less common in the dataset.

Figure 16:



Step C: Exploratory Data Analysis (EDA) and Visualization using Python

In this step we are using python to explore some parts of data. The figure 17 shows a world map titled "Countries with

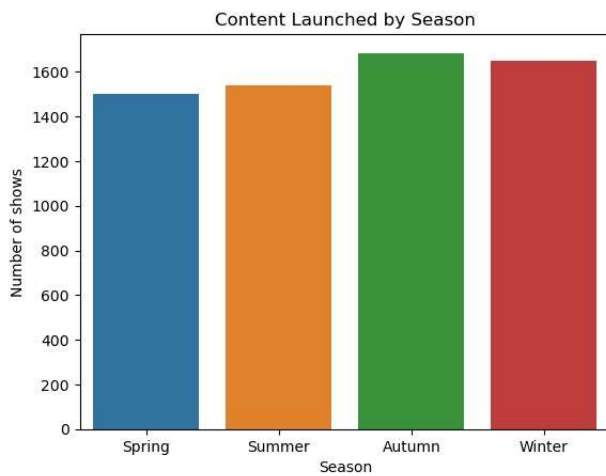
the Most Netflix Content," indicating the volume of Netflix content available in various countries. The countries are colored higher amounts of content. The United States (USA), India, France, Canada, UK are shaded, suggesting they have the highest volume of Netflix content in this representation. Other countries are not shaded, represents lesser content availability than darker shaded countries. This map provides a visual overview of the global distribution of Netflix's content, highlighting the disparities in content volume across different regions.

Figure 17:



The figure 18 shows a bar chart titled "Content Launched by Season," which compares the number of shows released across different seasons: Spring, Summer, Autumn, and Winter. The x-axis and y-axis represents the seasons of the year and number of shows. From the chart, it appears that the number of shows launched is relatively balanced across the seasons, with each bar showing a similar count ranging approximately between 1400 to 1600 shows. This suggests a consistent content release strategy throughout the year, rather than a concentration in any particular season.

Figure 18



Step D: SQL and AWS for Data Queries and Management

In the step D, Created a database using RDS and connected to the database using terminal and created a table similar to the Netflix dataset and loading the whole dataset from local to the table and displaying the schema of dataset as shown in figure 19 and 20.

Figure 19:

```
C:\Users\ASUS\OneDrive\Desktop>C:\Program Files\MySQL\MySQL Workbench 8.0 CE\mysql.exe" -u admin -h database-1.cxmyoqezn1.us-east-1.rds.amazonaws.com --local-infile -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 23
Server version: 8.0.35 Source distribution

Copyright (c) 2000, 2020, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
mysql> CREATE TABLE netflix_new (
  -> show_id text,
  -> type text,
  -> title text,
  -> director text,
  -> cast TEXT,
  -> country text,
  -> date_added DATE,
  -> release_year numeric,
  -> rating text,
  -> duration text,
  -> genres text,
  -> description text,
  -> month_added text,
  -> day numeric
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> Load data local infile 'C:/Users/ASUS/OneDrive/Documents/ait580data.txt' into table netflix_new fields terminated by '|' lines
terminated by '\n';
Query OK, 8793 rows affected, 31 warnings (0.41 sec)
Records: 8793 Deleted: 0 Skipped: 0 Warnings: 31
```

Figure 20:

```
mysql> desc netflix_new;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| show_id | text | YES | | NULL | |
| type | text | YES | | NULL | |
| title | text | YES | | NULL | |
| director | text | YES | | NULL | |
| cast | text | YES | | NULL | |
| country | text | YES | | NULL | |
| date_added | date | YES | | NULL | |
| release_year | decimal(10,0) | YES | | NULL | |
| rating | text | YES | | NULL | |
| duration | text | YES | | NULL | |
| genres | text | YES | | NULL | |
| description | text | YES | | NULL | |
| month_added | text | YES | | NULL | |
| day | decimal(10,0) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
14 rows in set (0.01 sec)
```

This query show in figure 21 retrieves the titles and durations of TV shows from the database where the type is 'TV Show' and the duration is 8 seasons or more. The output provides various TV shows alongside their respective durations, all of which have at least 8 seasons. Some of the TV show titles are "The Great British Baking Show," "My Little Pony: Friendship Is Magic," "American Horror Story," "Arrow," "Naruto," "Dexter," "The Office (U.S.)," and "Weeds."

Figure 21:

```
mysql> select title, duration from netflix_new where type='TV Show' and duration >= '8 Seasons';
```

title	duration
The Great British Baking Show	9 Seasons
Saved by the Bell	8 Seasons
My Little Pony: Friendship Is Magic	9 Seasons
Still Game	9 Seasons
American Horror Story	9 Seasons
Wentworth	8 Seasons
Call the Midwife	9 Seasons
Girlfriends	8 Seasons
Arrested	8 Seasons
The Ranch	8 Seasons
Trolls: The Beat Goes On!	8 Seasons
Harvey	9 Seasons
Spirit: Riding Free	8 Seasons
Vikings: Legendary Defender	8 Seasons
Under Arrest	8 Seasons
The Vampire Diaries	8 Seasons
Forensic Files	9 Seasons
Club Friday The Series 6	9 Seasons
Club Friday The Series 8	8 Seasons
Dexter	8 Seasons
Portlandia	8 Seasons
Royal Pains	8 Seasons
That '70s Show	8 Seasons
The Andy Griffith Show	8 Seasons
The Office (U.S.)	8 Seasons
Wings	8 Seasons

The query provided in figure 22 purpose is to count the number of shows (identified by show_id) added to the database for each month. The results are grouped by the 'month_added' column. According to the results, August had the highest number of shows added (769), while there are months with lower counts, like February (562).

Figure 22:

```
mysql> select count(show_id) , month_added from netflix_new group by month_added;
```

count(show_id)	month_added
1	month_added
769	September
754	August
827	July
728	June
632	May
763	April
740	March
562	February
737	January
811	December
785	November
760	October
4	NULL

14 rows in set (0.02 sec)

```
mysql>
```

VI. DISCUSSION

The study presented in this document offers profound insights into the evolution of Netflix's content distribution strategies, the detection of patterns based on textual data, and the increased focus from films to television series.

a. Evolution of content distribution across countries over time:

The research explains that Netflix has been expanding its presence globally, not only targeting English-speaking nations but also diversifying the markets such as India, Brazil, and South Korea. This expansion strategy aligns with Netflix's goal to attract a worldwide audience by investing in regional content, aiding in adherence to local content requirements. While the United States continues to be a major market, the

rise in international regions suggests a customized approach to content that aligns with local cultural tastes and preferences.

b. Patterns of similarity across movies and TV series based on text attributes:

Utilizing sophisticated NLP technologies like TF-IDF and cosine similarity matrices, the analysis revealed how Netflix strategically curates content with thematic and stylistic resemblances. This method probably helps in fostering an engaging user experience by suggesting titles that viewers might find similar to their preferences. The identified clusters may reflect Netflix's tactic of grouping content that caters to specific niche audiences, thus improving the delivery of personalized content.

c. Trends in the emphasis on developing TV series versus movies:

The analysis concerning the focus on TV series over films uncovers a notable strategic pivot. In recent times, there has been a growing emphasis on TV series, presumably because they engage subscribers for a longer duration than films, which generally have shorter engagement times. This trend mirrors viewer preferences for serialized content, which allows for more complex narratives and character development, attributes that are increasingly appreciated in the contemporary entertainment scene.

VII. LIMITATIONS

While this research offers valuable insights into Netflix's content strategy, it faces several constraints. The dataset used may not completely capture the diversity of global content because of inherent biases or differences in regional access, which could restrict the broader applicability of the results. Additionally, the dependence on particular algorithms and technologies for analyzing text might overlook finer linguistic details, thus limiting the thoroughness of textual analysis. The study also tends to ignore specific demographic preferences by concentrating on wider trends. Future studies would benefit from utilizing a larger, regularly updated dataset, employing more sophisticated natural language processing tools, and integrating predictive models to deepen the understanding of trends in streaming media and viewer behaviors.

VIII. FUTURE WORK

Further studies could investigate how variations in content preferences across different regions affect both the growth and retention of Netflix subscribers and the effectiveness of the company's content policies. Long-term research might also reveal the effects of shifting consumer patterns on Netflix's

future content approaches. Additionally, employing machine learning to forecast content success based on pre-release characteristics could significantly aid Netflix in refining its content development and procurement strategies.

Research into the consequences of Netflix's investments in original programming on its financial outcomes and brand image could prove to be informative. Moreover, examining the environmental consequences of streaming services and the measures companies like Netflix take to incorporate sustainability into their operational and content strategies could provide important findings.

IX. CONCLUSION

This research offers an in-depth evaluation of Netflix's approach to content, identifying key patterns in how content is distributed, the resemblances among various content forms, and a noticeable pivot towards increasing TV series production. The results indicate that Netflix is adeptly aligning with international trends and the preferences of viewers, a crucial factor for its ongoing development and competitive standing in the streaming industry.

The study emphasizes the critical role of employing data analytics and sophisticated text analysis methods to predict and comprehend consumer behaviors. As streaming platforms evolve, such knowledge will be pivotal for Netflix and similar entities to preserve their market lead and effectively serve a diverse, changing worldwide audience.

In pursuing these strategies, Netflix not only improves its service offerings but also contributes significantly to the wider conversation on digital media consumption, establishing standards and recommended practices for the industry.

X. REFERENCES

- [1] K. Chatterjee, "The Rise of Streaming Platforms: A Revolution in Entertainment," Medium, Jun. 12, 2023. <https://medium.com/@kasturichatterjee1108/the-rise-of-streaming-platforms-a-revolution-in-entertainment-3553b094d799>
- [2] A. D. Lotz, O. Eklund, and S. Soroka, "Netflix, library analysis, and globalization: Rethinking mass media flows," *Journal of Communication*, vol. 72, no. 4, pp. 511–521, Jun. 2022, doi: <https://doi.org/10.1093/joc/jqac020>.
- [3] J. A. Starosta and B. Izydorczyk, "Understanding the Phenomenon of Binge-Watching—A Systematic Review," *International Journal of Environmental Research and Public Health*, vol. 17, no. 12, p. 4469, Jun. 2020, doi: <https://doi.org/10.3390/ijerph17124469>.
- [4] "International Journal of Media & Cultural Politics," Intellect Books. <https://www.intellectbooks.com/international-journal-of-media-cultural-politics>
- [5] V. K. Adhikari et al., "Measurement Study of Netflix, Hulu, and a Tale of Three CDNs," in *IEEE/ACM Transactions on Networking*, vol. 23, no. 6, pp. 1984–1997, Dec. 2015, doi: 10.1109/TNET.2014.2354262.
- [6] N. Kamarudin, A. Daheche and A. Khmag, "Netflix User and Movies Interest Analysis for Asian Countries," 2022 9th International Conference on Electrical and Electronics Engineering (ICEEE), Alanya, Turkey, 2022, pp. 339–343, doi: 10.1109/ICEEE55327.2022.9772585
- [7] Devashree, H. Goel, N. Sharma and M. Mangla, "Analysis and Visualization of Netflix Shows," 2022 4th International Conference on Artificial Intelligence and Speech Technology (AIST), Delhi, India, 2022, pp. 1–6, doi: 10.1109/AIST55798.2022.10065331
- [8] "Netflix-assignment1 - dataset by vpasupu," data.world. <https://data.world/vpasupu/netflix-assignment1> (accessed Apr. 28, 2024).
- [9] NLTK, "Natural Language Toolkit — NLTK 3.4.4 documentation," Nltk.org, 2009. <https://www.nltk.org/>
- [10] "R Documentation and manuals | R Documentation," Rdocumentation.org, 2019. <https://www.rdocumentation.org/>
- [11] Amazon, "AWS Documentation," Amazon.com, 2019. <https://docs.aws.amazon.com/>
- [12] Achar Bhargav, Vybhav & Choi, Seongwoo & Haddad, David. (2022). *Data Analysis on Netflix datasets*.
- [13] D. Goel and D. Batra, "Predicting User Preference for Movies using NetFlix database." Accessed: Apr. 29, 2024. [Online]. Available: https://faculty.cc.gatech.edu/~dbatra/papers/goel_batra_netflix.pdf
- [14] S. Lall and R. Sivakumar, "A Real-world Dataset of Netflix Videos and User Watch-Behavior: Analysis and Insights," ICC 2021 - IEEE International Conference on Communications, Montreal, QC, Canada, 2021, pp. 1–7, doi: 10.1109/ICC42927.2021.9500669.

[15] K. Vadloori and S. Sanghishetty, "Exploratory and Sentiment Analysis of Netflix Data Netflix Data -which is an Open-Source Data Set obtained from Kaggle -that was wrangled and exercised to derive maximum insights using EDA -Exploratory Data Analysis and Sentiment Analysis after the amalgamation of two additional sets - Geographical Latitudes & Longitudes and Netflix Title Critics/Reviews Data Set. The project is made using different utility analytical tools present in Python Library of versatile packages. This paper introduces systematic and insightful usage of methods for Exploratory Data Analysis & Sentiment Analysis by utilizing various packages concerned." Accessed: Apr. 29, 2024. [Online]. Available: <https://www.ijert.org/research/exploratory-and-sentiment-analysis-of-netflix-data-IJERTV10IS090123.pdf>